

The Break-Even Externality

Prices vs. Quantities Revisited — DiPasquale, Glaeser, Guren, Willen

For a given subsidy s , how big must the externality be to raise welfare?

$$X^* = s \left(\frac{1}{2} + \lambda \Gamma + (1 - \delta) \kappa \left(\Gamma - \frac{1}{2} \right) \right) \quad \text{where} \quad \Gamma \equiv \frac{q(s)}{\Delta_Q} = \frac{\text{total units}}{\text{marginal units}} \geq 1$$

Policy	Horizon	s per unit	$s\Gamma$ per marginal unit	Γ	Break-even externality X^* , $\lambda = 0$, $\kappa = 1$			
					$\delta = 1$	$\delta = 0.75$	$\delta = 0.5$	$\delta = 0$
					per unit, NPV \$000s			
100% of revenue	∞	190	549	2.9	95	209	322	549
100% of revenue	29	111	427	3.9	55	148	241	427
100% of revenue	15	69	371	5.4	35	119	203	371
75% of revenue	29	83	415	5.0	42	135	228	414
\$5000/unit	29	79	345	4.4	40	116	192	345
\$2500/unit	29	46	299	6.5	23	92	161	299

δ , κ , λ : definitions and values used

- δ — **weight on developer profit** in the welfare function
 - $\delta = 1$ counts profit in full, $\delta = 0$ counts it zero
- $\kappa \equiv C_2 / (C_2 - B_2^I)$ — **share of s that raises the producer price**
 - needs B_2^I , C_2 separately; only $D = C_2 - B_2^I = s / \Delta_Q$ is estimated
 - set to $\kappa = 1$ in every column; $\kappa = 1 \iff B_2^I = 0$ (flat demand)
 - enters only as $(1 - \delta)\kappa$
- λ — marginal cost of public funds. $\lambda = 0$ in every column; $\lambda > 0$ adds $\lambda s \Gamma$
- $\Delta_Q = 1,786$ units; Boston stock $\approx 300,000$

The columns

- s — net cost $\div$ **total** units permitted
- $s\Gamma$ — net cost $\div$ **marginal** units = \$549k = Table 4 col. (7)
- X^* — **per unit**, not per marginal unit
- $\delta = 1$: bracket = $\frac{1}{2}$, so $X^* = \frac{1}{2}s = \$95\text{k}$. **Γ absent**
- $\delta = 0$, $\kappa = 1$: bracket = Γ , so $X^* = s\Gamma$. **Last column = column 2**
- Range \$95k to \$549k; ratio = $\Gamma = 2.9$

All entries at $\lambda = 0$ and $\kappa = 1$, so the bracket is $\frac{1}{2} + (1 - \delta)(\Gamma - \frac{1}{2})$, which equals $\frac{1}{2}$ at $\delta = 1$ and Γ at $\delta = 0$. $\kappa = 1$ is an upper bound; any $\kappa < 1$ moves every entry toward the $\delta = 1$ column. Costs in NPV \$000s per unit; s is net cost over *total* units permitted, $s\Gamma$ is net cost over *marginal* units (Table 4 column (7)). Source: Programs/06e.slide.tables.R, which derives every column from Table 4 and asserts both ends of the sweep — $X^* = \frac{1}{2}s$ at $\delta = 1$ and $X^* = s\Gamma$ at $\delta = 0$ — before writing.

The Break-Even Externality: Sweeping λ at $\delta = 1$

Page 1: $\lambda = 0$, δ swept. This page: $\delta = 1$, λ swept.

At $\delta = 1$ the κ term drops out:

$$X^* = s\left(\frac{1}{2} + \lambda\Gamma\right) \quad \text{so } X^* = s\Gamma \iff \lambda^* = 1 - \frac{1}{2\Gamma}$$

Policy	Horizon	s per unit	$s\Gamma$ per marginal unit	Γ	Break-even externality X^* , $\delta = 1$, κ irrelevant				
					$\lambda = 0$	$\lambda = 0.2$	$\lambda = 0.3$	$\lambda = 0.5$	λ^* where $X^* = s\Gamma$
100% of revenue	∞	190	549	2.9	95	205	260	370	0.83
100% of revenue	29	111	427	3.9	55	141	184	269	0.87
100% of revenue	15	69	371	5.4	35	109	146	220	0.91
75% of revenue	29	83	415	5.0	42	124	166	249	0.90
\$5000/unit	29	79	345	4.4	40	109	143	212	0.89
\$2500/unit	29	46	299	6.5	23	83	113	172	0.92

This page vs. page 1

- Page 1: $\lambda = 0$, $\kappa = 1$, δ swept $1 \rightarrow 0$
- This page: $\delta = 1$, λ swept $0 \rightarrow 0.5$
 - κ multiplied by $1 - \delta = 0$, so it **drops out**
- Page 1's bracket $\frac{1}{2} + (1 - \delta)(\Gamma - \frac{1}{2})$ is **bounded**: $\frac{1}{2}$ to Γ
- This bracket $\frac{1}{2} + \lambda\Gamma$ is **unbounded** in λ
- $X^* = s\Gamma$ at $\lambda^* = 1 - 1/(2\Gamma)$ — last column, **0.83 to 0.92**
 - λ^* depends on Γ only, not on s
 - $\partial\lambda^*/\partial\Gamma > 0$

The two channels compared

- $\lambda = 0$, $\delta = 1$: **\$95k** on both pages
- Row 1, holding the other parameter at its Pigou value:
 - $\delta = 0.5$, $\lambda = 0 \Rightarrow$ **\$322k**
 - $\lambda = 0.3$, $\delta = 1 \Rightarrow$ **\$260k**
- General formula carries both terms: $\lambda\Gamma + (1 - \delta)\kappa(\Gamma - \frac{1}{2})$
 - additive, so each page is a **lower bound** given the other parameter
- Γ multiplies both terms; absent from neither

All entries at $\delta = 1$, so the κ term is multiplied by $1 - \delta = 0$ and drops out; no assumption about pass-through is used on this page. λ^* solves $s(\frac{1}{2} + \lambda\Gamma) = s\Gamma$, giving $\lambda^* = 1 - 1/(2\Gamma)$, a function of Γ alone. Costs in NPV \$000s per unit; s is net cost over *total* units permitted, $s\Gamma$ is net cost over *marginal* units (Table 4 column (7)). Source: Programs/06e_slide_tables.R, which asserts $X^* = \frac{1}{2}s$ at $\lambda = 0$ and $X^* = s\Gamma$ at λ^* before writing.